Homework

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Recall the spatially bounded wave equation:

$$u_{tt} = u_{xx}, \qquad 0 < x < l, \quad t > 0, \tag{1}$$

and let $F(t) = \int_0^t f(\tau)\, d\tau$. If $u(0,t) = f(t)$ and $u(l,t) = 0$, then the solution is given below:

$$u(x,t) = \sum_{n=0}^{\infty} f(t - x - 2nl) + \sum_{n=1}^{\infty} f(t + x - 2nl). \tag{2}$$

1. Suppose $u(0,t) = f(t)$ and $u_x(l,t) = 0$. Then, we obtain the following solutions:

$$\begin{aligned} u(x,t) &= f(t-x), && 0 \le t < l \\ u(x,t) &= f(t-x) + f(t+x-2l), && l \le t < 2l \\ u(x,t) &= f(t-x) + f(t+x-2l) - f(t-x-2l), && 2l \le t < 3l. \end{aligned}$$

Proof (Hernandez). When $3l \le t < 4l$, we obtain the following:

$$\begin{aligned} u(x,t) &= f(t-x) + f(t+x-2l) - f(t-x-2l) - f(t+x-4l) \\ u_x(l,t) &= f'(t-l) + f'(t-l) + f'(t-3l) - f'(t-3l) = 0, \end{aligned}$$

which leads to the following solution:

$$u(x,t) = \sum_{n=0}^{\infty} (-1)^n f(t - x - 2nl) + \sum_{n=1}^{\infty} (-1)^{n+1} f(t + x - 2nl).$$

□

2. Suppose $u_x(0,t) = f(t)$ and $u(l,t) = 0$. Then, $u(x,t) = -F(t-x)$ if $x > 0$.

Proof (Hernandez). We compute the solution on some intervals below:

$$\begin{aligned} u(x,t) &= -F(t-x), && 0 \le t < l \\ u(x,t) &= -F(t-x) + F(t+x-2l), && l \le t < 2l \\ u(x,t) &= -F(t-x) + F(t+x-2l) + F(t-x-2l), && 2l \le t < 3l, \end{aligned}$$

and for all $t > 0$ below:

$$u(x,t) = \sum_{n=0}^{\infty} -F(t - x - 2nl) - \sum_{n=1}^{\infty} F(t + x - 2nl).$$

□

3. Suppose $u(0,t) = 0$ and $u(l,t) = f(t)$. A change of variables reveals the following solution:

Proof (Hernandez).

$$u(x,t) = \sum_{n=0}^{\infty} f(t+x-2nl) + \sum_{n=1}^{\infty} f(t-x-2nl).$$

□

4. Suppose $u(0,t) = f(t)$ and $u(l,t) = g(t)$. The solution is given by super-imposing the solution v where $g(t) = 0$ and w to the problem above where $f(t) = 0$.